
An Integrated Model of Training Evaluation and Effectiveness

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A decade of training evaluation and training effectiveness research was reviewed to construct an integrated model of training evaluation and effectiveness. This model integrates four prior evaluation models and results of 10 years of training effectiveness research. It is the first to be constructed using a set of strict criteria and to investigate the evaluation and effectiveness relationships with an evaluation measure proposed several years ago, posttraining attitudes. Evaluation measures found to be related to posttraining attitudes were cognitive learning, training performance, and transfer performance. Training effectiveness variables found to be related to posttraining attitudes were pretraining self-efficacy, experience, posttraining mastery orientation, learning principles, and post-training interventions. Overall, 10 training effectiveness variables were found to consistently influence training outcomes. Results also reveal that reaction measures and training motivation are two areas needing further development and research. These findings as well as other areas requiring research attention are discussed.

Keywords: *training; training evaluation; training effectiveness; post-training attitudes; training motivation; organizational characteristics; training characteristics; individual characteristics*

Among the many recent contributions to the training literature, training evaluation and training effectiveness have received considerable attention (e.g., Broad & Newstrom, 1992; Goldstein, 1991; Holton, 2003; Holton &

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Baldwin, 2000; Kraiger, 2002; Noe & Colquitt, 2002; Torres & Preskill, 2001). In particular, expansions of Kirkpatrick's (1976) four-level evaluation strategy (Holton, 1996; Kraiger, 2002; Kraiger, Ford, & Salas, 1993; Tannenbaum, Cannon-Bowers, Salas, & Mathieu, 1993) and comprehensive conceptual models of training effectiveness (Holton, 1996; Tannenbaum et al., 1993) have emerged as important works. However, one evaluation measure proposed more than 10 years ago, posttraining attitudes, has not been sufficiently incorporated into evaluation models. In addition, a comprehensive update of all the variables now believed to contribute to training effectiveness has not been done for several years. As a result, the purpose of this article is to review the past decade of research on training evaluation and effectiveness and summarize these findings with an integrated model of training evaluation and effectiveness (IMTEE). This article will begin by clarifying the distinction between training evaluation and training effectiveness. This will be followed by a review of prior evaluation models and effectiveness models found in the literature. Finally, the IMTEE, which is based on the past 10 years of research, will be presented and areas for further research will be identified.

Evaluation Versus Effectiveness

Training evaluation and training effectiveness are sometimes used interchangeably; however, they are two separate constructs. A practical example may help to clarify their differences. Recently, a government employment agency was under a court order to redesign and administer selection tests for more than 30 jobs. The deadline required that the staff work numerous hours of overtime for several months in a row. A year into the project a number of employees had left the agency and several employees were calling in sick on a regular basis. To prevent further employment loss and provide assistance to employees during the remainder of the project, the agency started a training program for dealing with burnout. All supervisors attended training and asked their subordinates to also attend. The training program was designed to include lecture, humor, and practice of various stress-reducing techniques. The program also engaged trainees in conversation geared to enhance coworker and supervisor support after training. Finally, the supervisors were encouraged to discuss their own continued practice of stress-reduction techniques with their subordinates.

Training evaluation. In the above example, the agency had two purposes for implementing burnout training: to assist employees with burnout and reduce employment loss. To examine if training assisted employees, the agency chose posttraining attitude and transfer evaluation measures. Specifically, measures of self-efficacy for dealing with stressful circumstances were administered

before and after training to see if there was an increase in self-efficacy after training. A self-report transfer measure was administered 2 weeks after training to determine if employees were using the techniques learned in training. To evaluate whether training reduced employment loss, the agency calculated turnover and the number of absences for 3 months before and after training and compared the results. Finally, to evaluate the appropriateness of training content and design, the agency used reaction measures to evaluate the relevance and usefulness of the training program. Thus, training evaluation is a measurement technique that examines the extent to which training programs meet the goals intended. The evaluation measures used depend on those goals and can include evaluation of training content and design, changes in learners, and organizational payoffs.

Training effectiveness. In addition to evaluating training results, the above employment agency makes a concerted effort to examine the particular aspects of the environment, training program, and employees that make the programs successful or unsuccessful. Although a scientific study of training effectiveness was not done in this case (e.g., developing two training programs with differing characteristics and comparing the results of each), effectiveness could still be measured through the posttraining attitude and transfer measures. Recall that the burnout training intervention paid particular attention to supervisor support, coworker support, humor, and practice. The transfer measure that was administered 2 weeks after training included items to ascertain the effect of these characteristics. Responses from the items of these scales as well as the self-efficacy measures were then compared to results (i.e., employee assistance and employment loss) to determine which, if any, of the characteristics were effective or not effective in producing change. Through these means the agency was able to improve future training programs. Simply stated, training effectiveness is the study of the variables that likely influence training outcomes at different stages (i.e., before, during, and after) of the training process. These effectiveness variables have the potential to increase or decrease the likelihood of successful training outcomes and are typically studied in three broad categories: individual, training, and organizational characteristics.

In summary, training evaluation is a methodological approach for measuring learning outcomes. Training effectiveness is a theoretical approach for understanding those outcomes. Because training evaluation focuses solely on learning outcomes, it provides a microview of training results. Conversely, training effectiveness focuses on the learning system as a whole, thus providing a macroview of training outcomes. Evaluation seeks to find the benefits of training to individuals in the form of learning and enhanced on-the-job performance. Effectiveness seeks to benefit the organization by determining why individuals learned or did not learn. Finally, evaluation results describe what happened as a result of the training inter-

vention. Effectiveness findings tell us why those results happened and so assist experts with developing prescriptions for improving training.

Training Evaluation Models

As previously mentioned, training evaluation is the measurement of a training program's success or failure with regard to content and design, changes in learners, and organizational payoffs. The evaluation techniques used to assess these depend on the evaluation model chosen, as four different models have been proposed. The first model, Kirkpatrick's four-dimensional measurement typology (i.e., reactions, learning, behavior, results), is perhaps the simplest method for understanding training evaluation and the most frequently cited technique. In this model, learning is measured during training and refers to attitudinal, cognitive, and behavioral learning. Behavior refers to on-the-job performance and, thus, is measured after training. Additionally, reactions to training are related to learning, learning is related to behavior, and behavior is related to results.

In the second model, Tannenbaum et al. (1993) expanded on Kirkpatrick's typology by adding posttraining attitudes and dividing behavior into two outcomes for evaluation: training performance and transfer performance. In their model, reactions to training and posttraining attitudes are not related to any other target of evaluation. However, learning is related to training performance, training performance is related to transfer performance, and transfer performance is related to results.

In the third evaluation strategy, Holton (1996) included three evaluation targets: learning, transfer, and results. Reactions are not a part of Holton's model because reactions are not considered a primary outcome of training; rather, reactions are defined as a mediating and/or moderating variable between trainees' motivation to learn and actual learning. In this model, learning is related to transfer and transfer is related to results. In addition, Holton argued for an integration of evaluation and effectiveness. As a result, in his model particular effectiveness variables are outlined as important for measurement when evaluating training outcomes. The effectiveness aspects of Holton's model will be described in more detail in the section on training effectiveness models below.

The fourth and final evaluation strategy was provided by Kraiger (2002). This model emphasizes three multidimensional target areas for evaluation: training content and design (i.e., design, delivery, and validity of training), changes in learners (i.e., affective, cognitive, and behavioral) and organizational payoffs (i.e., transfer climate, job performance, and results). Reactions are considered a measurement technique for determining how effective training content and design were for the tasks to be learned. Kraiger asserted that reaction measures are not related to changes in learners or

organizational payoffs but that changes in learners are related to organizational payoffs.

Training Effectiveness Models

Training effectiveness is the study of the individual, training, and organizational characteristics that influence the training process before, during, and after training. Training needs analysis is recognized as one of the first important “before” contributions to training effectiveness (Salas & Cannon-Bowers, 2001). Although a full description is beyond the scope of this article, basically, a thorough needs analysis takes into account the individual differences of trainees, the organizational climate and objectives, and the characteristics of the task(s) to be learned. This information is then used to determine both the method and content of training. In sum, training cannot be effective unless it meets the individual, organizational, and task needs as identified by needs analysis.

Additional contributions to training effectiveness are three sets of characteristics. The first set is individual characteristics or the factors that trainees bring to the situation. These include personality traits, attitudes, abilities, demographics, experience, and expectations. Individual characteristics also include attitudinal constructs that were manipulated in training such as self-efficacy, goal orientation, and motivation. The second set of characteristics covers the context in which training is implemented or the organizational and situational characteristics. These include the organization’s climate for learning, history, policies, trainee selection technique, and trainee notification process. The final set is training characteristics, which includes aspects of the training program such as instructional style, practice, and feedback (Cannon-Bowers, Salas, Tannenbaum, & Mathieu, 1995; Tannenbaum et al., 1993).

Training experts typically study training effectiveness variables through the targets of evaluation. For example, the employment agency previously described assessed how self-efficacy, practice, humor, and supervisor and coworker support were related to changes in learners and organizational payoffs. Four of the five effectiveness models found in the literature focus primarily on one evaluation measure, transfer performance. These effectiveness models focus on the relationship between learning as a whole (i.e., attitudes, cognitive, and behavioral) and transfer performance and provide insight into how the three sets of characteristics are related to learning and transfer performance. For example, Baldwin and Ford’s (1988) model suggests that individual and organizational characteristics are directly related to learning and transfer performance, whereas all three sets of characteristics have an indirect relationship with transfer performance through learning. Although not present in their model, Baldwin and Ford also suggested

that individual characteristics are related to both training and organizational characteristics. These relationships also result in increases (or decreases) in learning and transfer performance. This model was extended by Holton and Baldwin's (2000) training effectiveness model, which more explicitly identifies particular characteristics affecting learning and transfer performance. These characteristics include ability, motivation, individual differences, prior experience with the transfer system, learner and organizational interventions (e.g., preparation, supports), and training content and design.

Holton's (1996) model of training effectiveness also has particular individual, training, and organizational characteristics as primary or secondary underlying variables that influence the outcomes of training. Overall, Holton's model suggests that the three sets of characteristics are directly related to learning and transfer performance. However, there are also indirect relationships because of interactions between the characteristics. For example, Holton suggested that a primary individual characteristic, motivation, interacts with training and organizational characteristics, thus influencing the outcomes of training. Although Holton provided useful guidelines for measuring training effectiveness, few studies (e.g., Holton, 2003; Holton, Bates, & Ruona, 2000) have simultaneously measured the various aspects suggested by the author. These authors developed the Learning Transfer System Inventory with the effectiveness variables outlined in the model and found support for the model's construction.

The fourth training effectiveness model is not a model per se; however, it clarifies the process nature of training effectiveness. That is, to enhance training outcomes, Broad and Newstrom (1992) prescribed strategies that organizations can implement before, during, and after training. The authors suggested that all three characteristics (i.e., individual, training, and organizational) are related to learning and transfer performance.

The fifth and final training effectiveness model described here displayed relationships between the three characteristic types and four evaluation targets: cognitive learning, training and transfer performance, and results. In this model, Tannenbaum et al. (1993) suggested that individual and training characteristics are directly related to cognitive learning and training performance, whereas individual and organizational characteristics are directly related to transfer performance. This model also outlines underlying interactions between the three sets of characteristics. For example, similar to Baldwin and Ford (1988) and Holton (1996), organizational, individual, and training characteristics are posited to influence trainee motivation, which in turn is related to cognitive learning and transfer performance.

The fact that the above discussion of training effectiveness could not be presented without mentioning evaluation measures demonstrates that, although training evaluation and training effectiveness are distinct concepts, they are also necessarily related. Therefore, models that fully inte-

grate both concepts provide better pictures of their interrelations and help with understanding each individual concept better than nonintegrated models of the concepts on their own. This article will propose such a model by integrating the past 10 years of training evaluation and effectiveness research. The method used, the results, and a comparison between the IMTEE and prior models will be the topics of the following sections.

The IMTEE

To review and integrate the literature on training evaluation and effectiveness, a literature search was conducted for empirical investigations of adult training that were published during the 10 years prior to this writing (i.e., 1993 to 2002). The search included empirical studies published from the following journals: *Applied Psychology: An International Review*, *Computers in Human Behavior*, *Ergonomics*, *Evaluation and the Health Professions*, *Human Factors*, *Human Resource Development Quarterly*, *International Journal of Human-Computer Studies*, *Journal of Applied Psychology*, *Journal of Educational Computing Research*, *Journal of Experimental Psychology: Human Perception and Performance*, *Journal of Instructional Psychology*, *Journal of Management*, *Journal of Occupational & Organizational Psychology*, *Journal of Organizational Behavior*, *Journal of Organizational Behavior Management*, *Military Psychology*, *Organizational Behavior & Human Decision Processes*, *Organizational Dynamics*, *Personnel Psychology*, and *Psychological Record*. The variables investigated in each article were examined, and studies that did not analyze relationships between the characteristics of interest (i.e., individual, organizational, training) and targets of evaluation (i.e., reactions, learning, transfer, etc.) were eliminated. This resulted in a total of 73 studies (see the appendix). Although there was some overlap, there were approximately 52 articles that examined individual characteristics, 16 articles that assessed training characteristics, 4 articles that investigated organizational/situational characteristics, and 1 article that examined the relationships between evaluation measures.

Because of the small number of studies investigating particular relationships, a statistical meta-analytic technique was not feasible. Assuming that generalization of training effectiveness should ideally be a result of replication and consistency (Aronson, Ellsworth, Carlsmith, & Gonzales, 1990; Cook & Campbell, 1979), the IMTEE was constructed by including only those variables that met a strict set of criteria. To be included in the model, a minimum of three studies finding a significant ($p < .05$) relationship in the same direction (i.e., positive or negative) was required. In addition, if any variable resulted in consistently mixed relationships across studies (i.e., negative, no findings, and positive), the characteristic was not included in

the model. However, because of the differing nature of research across posttraining attitudes as well as individual, training, and organizational characteristics, some flexibility in classifying each set of variables was necessary. The following sections explain how individual, training, and organizational characteristics were categorized as well as the solution adopted when applying the criteria to posttraining attitudes.

Individual Characteristics

An initial goal of this research effort was to identify the specific variables within each category of characteristics that contribute to training effectiveness. This was relatively unproblematic with all but one individual characteristic: motivation. The plethora of motivation types investigated by researchers hampered a simple classification of results. Tannenbaum et al. (1993) alone included four types of motivation in his model of training effectiveness, Holton (1996) included two, and the literature review revealed seven methods for measuring motivation, each involving a different motivational aspect. Furthermore, some studies combined different aspects of motivation into one overall motivational scale. Therefore, a decision was made to combine all studies of motivation to determine inclusion eligibility because it was difficult to separate the effects of the differing scales. It is acknowledged that this simplicity in the model belies a greater complexity between the variables.

Training Characteristics

Training characteristics research is in a more advanced stage than that of individual and organizational characteristics. It appears that researchers have moved beyond investigating the effectiveness of particular instructional techniques (e.g., lectures, role-plays, group exercises, videos, games, etc.) and learning principles (e.g., behavior modeling, practice, use of identical elements, part- vs. whole-task training, feedback, etc.). Instead, experts are now attempting to identify packages of these characteristics (e.g., group exercises, practice, identical elements, and feedback) most effective for learning specific skills (e.g., team building). In addition, new instructional techniques such as error training are also being investigated.

The results of this review, as well as Tannenbaum et al.'s (1993) work, suggest that the effectiveness of a particular instructional technique depends on the content of training. Therefore, instructional techniques were not included in this investigation. On the other hand, learning principles have been found to contribute to learning and transfer performance in a wide variety of instructional environments (Tannenbaum et al., 1993). Although there were not enough studies to investigate each learning principle with the criteria set forth in this investigation, there were enough studies to treat the

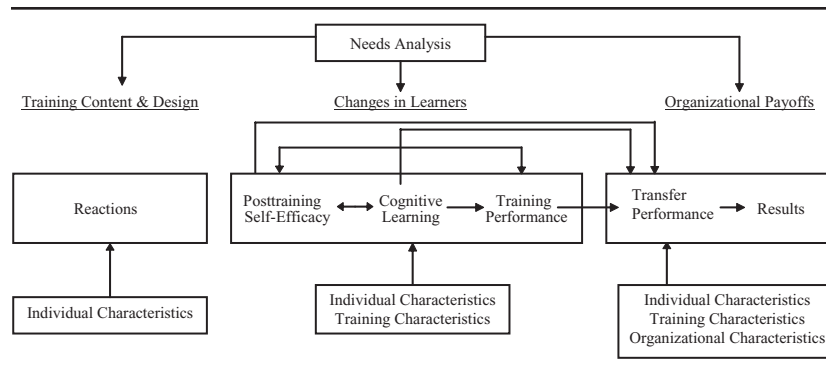


FIGURE 1: Integrated Model of Training Evaluation and Effectiveness

group of learning principles as one variable and subsequently examine the group for inclusion in the model.

Organizational Characteristics

Organizational characteristics have been less thoroughly investigated than individual and training characteristics resulting, first, in a small number of articles (i.e., four) found in the literature review. In addition, researchers have yet to agree on a consistent method for measuring organizational climate. The four studies combined investigated a total of 22 facets of the work environment. Furthermore, three of these four studies investigated the relationships between aggregated scores from multidimensional organizational climate scales and the targets of evaluation, further thwarting our ability to examine the unique effects of specific organizational characteristics on training outcomes. Therefore, organizational characteristics as a whole were considered one characteristic termed *positive transfer environment* and subjected to the criteria for inclusion in the model outlined below.

Posttraining Attitudes

Posttraining attitudes include constructs such as self-efficacy, mastery orientation, organizational commitment, motivation, attitudes toward diversity, attitudes toward teamwork, and so on (Kraiger, 2002; Tannenbaum et al., 1993). To our knowledge, the IMTEE presented in Figure 1 is the first model to propose relationships between posttraining attitudes and training effectiveness variables as well as relationships between posttraining attitudes and the other targets of evaluation. However, posttraining self-efficacy was the only attitude found to meet the criteria for inclusion in the model. Thus, to fully explore the possible relationships between posttraining attitudes and the three sets of characteristics as well as the other tar-

gets of evaluation, we chose to apply the criteria to the studies investigating posttraining self-efficacy and replace the term *posttraining attitudes* with *posttraining self-efficacy* in the IMTEE (see Figure 1). It is possible that if the literature review had been expanded over a longer time period, more posttraining attitudes might have met the criteria. In fact, a post hoc analysis, which will be described in a later section, revealed that recent research on mastery orientation may support the relationships proposed in the IMTEE. The remainder of this article will be devoted to describing the proposed IMTEE.

The IMTEE

As can be seen in Figure 1, the IMTEE has four levels. Starting from the top is needs analysis. Needs analysis was not subjected to the criteria for inclusion in the model; however, this variable was included because of its widely received association with training. The arrows from needs analysis to the second level of the model suggest that needs analysis contributes to the three overall targets of evaluation. Ideally, needs analysis results are used to develop training content and design that will enhance changes in learners and organizational payoffs. Thus, needs analysis is related to training content and design, changes in learners, and organizational payoffs (Kraiger, 2002; Tannenbaum et al., 1993).

The second and third levels of the IMTEE not only efficiently combine the four models of training evaluation but are also supported by recent research and theory. Kraiger's (2002) three areas of evaluation serve as umbrellas for the six evaluation measures proposed by the combined works of Kirkpatrick (1976), Holton (1996), and Tannenbaum et al. (1993). The six evaluation measures shown in the model and also proposed in Tannenbaum et al.'s work are reflective of the measures commonly used in current training research. With the exception of posttraining attitudes, the relationships shown between the evaluation measures are based on the theories proposed by all four evaluation models as well as a meta-analysis conducted by Alliger, Tannenbaum, Bennett, Traver, and Shotland (1997). The relationships between posttraining attitudes and the other evaluation measures were subjected to our criteria for inclusion in the model. However, because self-efficacy was the only posttraining attitude included in the model, the relationships between posttraining attitudes in general and other targets of evaluation may differ from what is shown in Figure 1.

Starting from left to right, training content and design can be evaluated by measuring reactions to training. Changes in learners can be assessed by measuring posttraining attitudes, cognitive learning, and training performance. Finally, organizational payoffs can be determined by measuring transfer performance and results. We are not suggesting that the methods for

evaluating training content and design, changes in learners, and organizational payoffs are limited to the six evaluation measures. The IMTEE merely reflects the current state of research. Kraiger et al. (1993) and Kraiger (2002) provided additional methods for evaluating training; however, few studies (e.g., Bell & Kozlowski, 2002) have shared their terminology or used the measurement suggestions offered by the authors. Each of the six targets of evaluation will now be defined and the relationships between them will be outlined.

Reactions. Moving from left to right, the first outcome is reactions to training and is typically a multidimensional construct (e.g., affective reactions, utility, relevance, etc.). Contrary to Kirkpatrick's (1976) suggestion that positive affective reactions are related to learning, Alliger et al.'s (1997) meta-analysis did not find a relationship between affective reactions and learning or any other training outcome. Likewise, three of the training evaluation models do not propose relationships between reactions and the other evaluation measures agree that positive reactions to training are not a predictor of learning (Holton, 1996; Kraiger, 2002; Tannenbaum et al., 1993). On the other hand, Alliger et al. (1997) did find that reactions regarding the usefulness of training programs were related to learning ($r = .26$) and transfer ($r = .18$). They further suggested that trainees may be the best source for determining the usefulness of training because they are familiar with their companies' organizational characteristics and know whether the skills being trained will be transferable to their jobs. Therefore, the IMTEE, as well as Alliger et al., Kirkpatrick, and Kraiger (2002), suggests that reactions (e.g., utility, relevance) be used to evaluate the appropriateness of training content and design and, consequently, to also assess the accuracy and thoroughness of needs analysis (see Figure 1). Future research is called for to verify this conclusion and elucidate methods for constructing reaction measures that are developed with training content and design in mind.

Posttraining attitudes. In Kirkpatrick's (1976) evaluation model, changes in attitudes as a result of training were subsumed under "learning." Both Tannenbaum et al. (1993) and Kraiger (2002) considered attitude change as an outcome deserving of independent attention in their evaluation models because a variety of possible affective outcomes of training exist (e.g., increases in self-efficacy, motivation, and organizational commitment, etc.). Furthermore, changes in attitudes are sometimes the purpose of training interventions (e.g., diversity training, attitudes toward teamwork, mastery goal orientation manipulations, etc.).

Given the criteria for inclusion in the model, three qualifying relationships were found between posttraining attitudes and other targets of evaluation. Posttraining self-efficacy is positively related to cognitive learning (Lorenz, Gregory, & Davis, 2000; Martocchio, 1994; Martocchio & Dulebohn, 1994; Martocchio & Judge, 1997), training performance (Cole & Latham, 1997; Davis, Fedor, Parsons, & Herold, 2000; Martocchio &

Dulebohn, 1994; Mathieu, Martineau, & Tannenbaum, 1993), and transfer performance (Ford, Smith, Weissbein, Gully, & Salas, 1998; Lorenz et al., 2000; Warr, Allan, & Birdi, 1999). Some studies reported that cognitive learning and training performance influenced posttraining attitudes, whereas others suggested that posttraining attitudes contributed to cognitive learning and training performance. Because a correlation coefficient does not elucidate the direction of a relationship, the IMTEE suggests that the relationship between cognitive learning and posttraining attitudes as well as training performance and posttraining attitudes may be reciprocal. Future research will be needed to verify this conclusion.

Another posttraining attitude that is frequently studied is mastery orientation. Although not meeting our criteria for inclusion in the model, like posttraining self-efficacy, posttraining mastery orientation was related to cognitive learning (Fisher & Ford, 1998; Hertenstein, 2001), training performance (Brett & VandeWalle, 1999), and transfer performance (Gist & Stevens, 1998; Krijger & Pol, 1995). It is hypothesized that future research will find mastery orientation as a variable that can be legitimately placed in the IMTEE as a posttraining attitude having the same relationships as posttraining self-efficacy.

This review found little research investigating changes in motivation as a result of training (e.g., Cole & Latham, 1997; Frayne & Geringer, 2000). These investigators used expectation measures of motivation and found a substantial increase in posttraining motivation. Several training effectiveness models have placed two different motivational aspects in their models as important effectiveness variables: motivation to learn and motivation to transfer (e.g., Baldwin & Ford, 1988; Holton, 1996; Holton & Baldwin, 2000; Tannenbaum et al., 1993). This strategy makes it difficult to examine how changes in motivation affect training outcomes. Furthermore, with this strategy, training experts cannot study aspects of training design and content or the organizational climate that may enhance motivation to learn or transfer. Consequently, there is a need for a method of measuring changes in motivation. One potential solution is to use a combined scale of motivation to learn and motivation to transfer such as the one recently developed and validated by Naquin and Holton (2002). In conclusion, the IMTEE suggests that posttraining attitudes should be a part of evaluation models because, across a variety of training programs, they are related to changes in learners and organizational payoffs.

Cognitive learning. Cognitive learning is the cognitive acquisition of knowledge and is typically measured through paper-and-pencil or electronically administered tests of information taught in training. Tannenbaum et al. (1993) further elaborated that the acquisition of knowledge can include increases in knowledge, a change in the mental structure of knowledge, or both. Kraiger (2002) expanded cognitive outcomes of training by emphasizing structural

knowledge, situated problem solving, ampliative skills, self-knowledge, and executive control. With regard to cognitive learning and other targets of evaluation, as previously discussed, there is a reciprocal relationship between post-training self-efficacy and cognitive learning. In addition, cognitive learning is related to training performance and transfer performance (Alliger et al., 1997; Kraiger et al., 1993; Tannenbaum et al., 1993).

Training performance. Training performance is the ability to perform a newly acquired skill at the end of training, prior to transfer, and is measured through observable demonstration that a trainee can implement the knowledge acquired in training. Kraiger (2002) proposed two dimensions of training performance: proceduralization (i.e., the ability to mimic modeled behaviors in training) and compilation (i.e., fluid performance with few errors after continued practice). Tannenbaum et al. (1993) reasoned that trainees may show competence in training but not transfer newly acquired skills to the work environment. As a result, training performance is likely to be greater than transfer performance (e.g., Ricci, Salas, & Cannon-Bowers, 1996). With regard to the relationships between training performance and other targets of evaluation, as previously mentioned, training performance is influenced by posttraining attitudes and cognitive learning. In addition, training performance positively influences posttraining attitudes and transfer performance (Alliger et al., 1997; Kirkpatrick, 1976; Kraiger, 2002; Tannenbaum et al., 1993).

Transfer performance. Transfer performance is behavioral changes on the job as a result of training and can be assessed via supervisor evaluations of on-the-job behavior or posttraining retests several months after training using the same or an alternate form of the training performance test. In their meta-analysis, Alliger et al. (1997) did not have sufficient studies assessing results of training to determine whether transfer performance is related to results. However, as stated by evaluation theorists (Kirkpatrick, 1976; Kraiger et al., 1993; Tannenbaum et al., 1993), the arrow from transfer performance to results in Figure 1 suggests that there is a positive relationship and a causal link between transfer performance and results. Theorists also suggest that this relationship is moderated by the quality of needs analysis. In other words, when training is aligned with the organizational objectives as determined by needs analysis, an increase in organizational effectiveness is likely to occur.

Results. The final dimension of training evaluation is results. This dimension refers to "quantifiable changes in related outcomes as a result of trainees' behavioral changes" (Tannenbaum et al., 1993, p. 24). For example, organizational benefits of transfer performance may include increased safety precautions, efficiency, morale, and quantity or quality of outputs (Holton, 1996; Kirkpatrick, 1976; Kraiger, 2002; Tannenbaum et al., 1993).

Our review of training evaluation did not find support for a hierarchical relationship between the evaluation measures. Indeed, relationships were

positive and causal; however, each outcome was not a sufficient predictor of the next. For example, Alliger et al.'s (1997) meta-analysis found that cognitive learning exhibited significant but small relationships with training performance ($r = .18$) and transfer performance ($r = .11$), and training performance demonstrated a significant but small relationship with transfer ($r = .18$). As can be seen by the small correlation coefficients found by these authors, outcome measures alone do not fully identify the constructs affecting training results. Holton (1996) explains it best: Without effectiveness variables, if evaluation research results in weak correlation coefficients, we do not know if the resulting outcomes are because of training design elements, individual trainee characteristics, or the organizational context. In short, training programs cannot be appropriately evaluated in isolation from their surrounding contexts (Tannenbaum et al., 1993). Thus, the following section describes the final level of the IMTEE: training effectiveness variables.

Moving from left to right, the results of our review found that individual characteristics are related to reactions; individual and training characteristics are related to all three measures of changes in learners; and individual, training, and organizational characteristics are related to transfer performance. To aid the following discussion, Table 1 presents the specific effectiveness variables found to affect training outcomes, the operational definitions of each effectiveness variable, and the studies finding significant relationships between those effectiveness variables and the outcomes of training.

Reactions. As can be seen in the second column of Table 1, two attitudinal, individual characteristics influence employees' reactions to training: pretraining self-efficacy and motivation (Cannon-Bowers et al., 1995; Facticeau, Dobbins, Russell, Ladd, & Kudisch, 1995; Quiñones, 1995; Tracey, Hinkin, Tannenbaum, & Mathieu, 2001; Warr et al., 1999; Warr & Bunce, 1995; Webster & Martocchio, 1995). Although not meeting the criteria for inclusion in the model, two studies found a negative relationship between maintenance interventions and reactions to training (Burke, 1997; Werner, O'Leary-Kelly, Baldwin, & Wexley, 1994). Note, however, that maintenance interventions were found to enhance transfer performance (Krijger & Pol, 1995; Morin & Latham, 2000; Stevens & Gist, 1997). These relationships suggest that even though trainees may not like maintenance interventions they are later benefited by increased on-the-job performance.

Posttraining attitudes. The results of this review found that changes in self-efficacy are related to individual characteristics such as pretraining self-efficacy, experience, and posttraining mastery orientation. Training characteristics related to posttraining self-efficacy are learning principles such as feedback, identical elements, and practice along with maintenance interventions involving visualization and mastery orientation manipulations

(text continues on page 404)

TABLE 1: Variable Definitions and Studies Finding Significant Relationships

Characteristic	Training Content and Design		Changes in Learners		Organizational Payoffs	
	Individual	Reactions	Posttraining Attitudes	Cognitive Learning	Training Performance	Transfer Performance
<i>Pretraining self-efficacy.</i> Self-report assessment prior to training where subjects rate the degree of confidence they have in their ability to perform a task at a specified level of proficiency.		Quiñones (1995); Tracey et al. (2001); Warr et al. (1999); Webster & Martocchio (1995)	Davis et al. (2000); Martocchio & Dulebohn (1994); Martocchio & Judge (1997); Mathieu et al. (1993); Saks (1995); Warr et al. (1999)	Bell & Kozlowski (2002); Cannon-Bowers et al. (1995); Tracey et al. (2001); Webster & Martocchio (1995)		Bell & Kozlowski (2002); Saks (1995); Warr et al. (1999)
<i>Pretraining motivation.</i> Self-report assessments of the degree trainees are motivated to attend the training program and/or their expectations of the outcomes of training once they complete training.		Cannon-Bowers et al. (1995); Fecteau et al. (1995); Quiñones (1995); Tracey et al. (2001); Warr et al. (1999); Warr & Bunce (1995)		Colquitt & Simmering (1998); Quiñones (1995); Tracey et al. (2001); Warr & Bunce (1995)		

(continued)

TABLE I (continued)

Characteristic	Training Content and Design		Changes in Learners		Organizational Payoffs	
	Individual	Reactions	Posttraining Attitudes	Cognitive Learning	Training Performance	Transfer Performance
<i>Experience.</i> Measured by (a) hours, months, and/or years of experience with specific tasks; (b) knowledge or work sample tests on the content of training materials prior to training; and/or (c) a scale asking whether participants have engaged in particular tasks in the past.			Davis et al. (2000); Lorenz et al. (2000); Warr et al. (1999)	Dyck & Smither (1996); Martocchio & Dulebohn (1994); Martocchio & Judge (1997)	Davis et al. (2000); Martocchio & Dulebohn (1994); Ree et al. (1995)	
<i>Posttraining mastery orientation.</i> Self-report measures of mastery orientation after training manipulation (e.g., training instructions remind trainees that ability is acquired, not fixed).			Ford et al. (1998); Martocchio (1994); Martocchio & Dulebohn (1994)			

Cognitive ability. Also referred to as intelligence or *g*, cognitive ability is an individual's capacity to learn, understand instructions, and solve problems. Tests of cognitive ability may include verbal and math questions; however, specific bodies of knowledge are also assessed and referred to as cognitive ability (e.g., spatial ability, electronics, science, mechanics, etc.).

Age. The age, in years and/or months, of trainees.

Bell & Kozlowski (2002); Cannon-Bowers et al. (1995); Carter (2002); Driskell et al. (1994); Dyck & Smither (1996); Ferguson et al. (2000); Fisher & Ford (1998); Gully et al. (2002); Hanisch & Hulin (1994); Hertenstein (2001); Martocchio & Judge (1997); Ree et al. (1995); Silva & White (1993); Werner et al. (1994)

Cannon-Bowers et al. (1995); Martocchio (1994); Warr & Bunce (1995)

(continued)

TABLE I (continued)

Characteristic	Training Content and Design		Changes in Learners		Organizational Payoffs	
	Individual	Reactions	Posttraining Attitudes	Cognitive Learning	Training Performance	Transfer Performance
<i>Learning principles.</i> Methods that are known to enhance learning across a wide variety of training tasks such as opportunities to practice, use of identical elements, behavior modeling, and providing feedback.			Ford et al. (1998); Karl et al. (1993); Warr et al. (1999)	Bell & Kozlowski (2002); May & Kahnweiler (2000); Moreland & Myaskovsky (2000); Simon & Werner (1996)	Bell & Kozlowski (2002); Goettl et al. (1996); Gopher et al. (1994); May & Kahnweiler (2000); Moreland & Myaskovsky (2000); Peck & Detweiler (2000); Simon & Werner (1996); Smith-Jentsch et al. (1996)	Bell & Kozlowski (2002); Gopher et al. (1994); Krijger & Pol (1995); Peck & Detweiler (2000)

<i>Posttraining interventions.</i> Interventions after training that are geared to facilitate skill transference to the work environment (e.g., mastery orientation manipulations, visualization, goal setting, etc.).	Gist & Stevens (1998); Martocchio (1994); Martocchio & Dulebohn (1994); Stevens & Gist (1997)	Burke (1997); Gist & Stevens (1998); Krijger & Pol (1995); Morin & Latham (2000); Richman-Hirsch (2001) Doane et al. (1996); Doane et al. (1999); Gist & Stevens (1998) Facticeau et al. (1995); Rouiller & Goldstein (1993); Rynes & Rosen (1995); Tracey et al. (1995)
<i>High difficulty.</i> A training manipulation; attendees learn tasks more difficult than those in the environment where skills will be transferred.		
<i>Positive transfer environment.</i> An environment that encourages trainees to implement skills learned in training such as supervisory support, rewards for applying learned skills, and follow-up performance evaluations.		

NOTE: All relationships are positive with the exception of age, which is negatively related to cognitive learning.

(Davis et al., 2000; Ford et al., 1998; Gist & Stevens, 1998; Karl, O'Leary-Kelly, & Martocchio, 1993; Lorenz et al., 2000; Martocchio, 1994; Martocchio & Dulebohn, 1994; Martocchio & Judge, 1997; Mathieu et al., 1993; Saks, 1995; Stevens & Gist, 1997; Warr et al., 1999).

Cognitive learning. Cognitive learning was related to pretraining self-efficacy and motivation, experience, and cognitive ability. In addition, age was found to be negatively related to cognitive learning; however, age was not related to training or transfer performance. Finally, learning principles such as practice, behavioral modeling, and providing feedback were also positively related to cognitive learning (Bell & Kozlowski, 2002; Cannon-Bowers et al., 1995; Carter, 2002; Colquitt & Simmering, 1998; Driskell, Hogan, Salas, & Hoskin, 1994; Dyck & Smither, 1996; Ferguson, Sanders, O'Hehir, & James, 2000; Fisher & Ford, 1998; Gully, Payne, Koles, & Whiteman, 2002; Hanisch & Hulin, 1994; Hertenstein, 2001; Martocchio, 1994; Martocchio & Dulebohn, 1994; Martocchio & Judge, 1997; May & Kahnweiler, 2000; Moreland & Myaskovsky, 2000; Quiñones, 1995; Ree, Carretta, & Teachout, 1995; Silva & White, 1993; Simon & Werner, 1996; Tracey et al., 2001; Warr & Bunce, 1995; Webster & Martocchio, 1995; Werner et al., 1994). Posttraining mastery orientation was an individual characteristic that did not meet the criteria because only two studies found a positive relationship between posttraining mastery orientation and cognitive learning (Fisher & Ford, 1998; Hertenstein, 2001).

Training performance. Based on our inclusion criteria, experience was the only individual characteristic related to training performance (Davis et al., 2000; Martocchio & Dulebohn, 1994; Ree et al., 1995). However, prior research suggests that, when motor skills are required for successful task accomplishment, physical ability and trainability may also be related to training performance (Tannenbaum et al., 1993). Training characteristics that were positively related to training performance included learning principles such as practice, behavior modeling, part- versus whole-task learning, and feedback (Bell & Kozlowski, 2002; Goettl, Yadrick, Connolly-Gomez, Regian, & Shebilske, 1996; Gopher, Weil, & Bareket, 1994; May & Kahnweiler, 2000; Moreland & Myaskovsky, 2000; Peck & Detweiler, 2000; Simon & Werner, 1996; Smith-Jentsch, Salas, & Baker, 1996).

Mixed relationships were found between cognitive ability and training performance. Six studies in this review found a positive relationship between cognitive ability and training performance (Bell & Kozlowski, 2002; Cannon-Bowers et al., 1995; Dyck & Smither, 1996; Eyring, Johnson, & Francis, 1993; Fisher & Ford, 1998; Gully et al., 2002), but four other studies did not (Driskell et al., 1994; Ree et al., 1995; Silva & White, 1993; Werner et al., 1994). These mixed results indicate that cognitive ability's relationship with training performance may be moderated by training content and is therefore necessary for performing some skills but not others. Future research is needed to clarify these results.

Inconsistent relationships between conscientiousness and training performance were also found. For example, one study resulted in a negative relationship (Stewart, Carson, & Cardy, 1996), two found a positive relationship (Cellar, Miller, Doverspike, & Klawnsky, 1996; Driskell et al., 1994), and one demonstrated no relationship (Herold, Davis, Fedor, & Parsons, 2002). On closer examination of the criteria, the relationships are understandable. For instance, one of the studies finding a positive relationship had performance aspects including uniform inspections, barracks demerits, and legal incidents (Driskell et al., 1994). Because conscientious individuals tend to follow rules, a positive relationship seems logical. On the other hand, the negative relationship between conscientiousness and performance found by Stewart et al. also seems reasonable. In this study, the employees were trained to violate organizational policies to satisfy customers. Conscientious individuals may find it more difficult to break the rules, even to help customers. In conclusion, it seems that the reason for mixed findings between conscientiousness and training performance is because of the nature of training content. Again, additional research can help to verify this conclusion.

Transfer performance. One individual characteristic was found to be related to transfer performance, pretraining self-efficacy. Under training characteristics, learning principles such as practice, part- versus whole-task learning, and feedback were positively associated with transfer performance. Posttraining maintenance interventions such as goal setting, visualization exercises, and mastery orientation manipulations were also related to transfer performance. The final training characteristic that was related to transfer performance is high difficulty, which is a training manipulation that requires trainees to learn tasks more difficult than those in the environment where skills will be transferred (Bell & Kozlowski, 2002; Burke, 1997; Doane, Alderton, Sohn, & Pellegrino, 1996; Doane, Sohn, & Schreiber, 1999; Fecteau et al., 1995; Gist & Stevens, 1998; Gopher et al., 1994; Krijger & Pol, 1995; Morin & Latham, 2000; Peck & Detweiler, 2000; Richman-Hirsch, 2001; Saks, 1995; Warr et al., 1999).

A positive transfer environment was also related to transfer performance (Fecteau et al., 1995; Rouiller & Goldstein, 1993; Rynes & Rosen, 1995; Tracey, Tannenbaum, & Kavanagh, 1995). As mentioned earlier, only one study (Rynes & Rosen, 1995) investigated relationships between unique factors of the transfer climate and training performance. This study reported relationships between transfer performance and 4 out of 12 dimensions of the environment (perceived supervisor support, mandatory attendance for managers, rewards for practicing skills, and follow-up evaluations beyond reaction measures), suggesting that individual organizational characteristics may be differentially related to training outcomes.

Two characteristics that did not meet the criteria for inclusion in the model were organizational commitment and posttraining mastery orienta-

tion, again, because only two studies found a positive relationship between these variables and transfer performance (Gist & Stevens, 1998; Krijger & Pol, 1995; Saks, 1995; Tesluk, Farr, Mathieu, & Vance, 1995). Future research should explore if these are indeed additional attitudinal variables that increase the effectiveness of training programs.

Training effectiveness is the study of the variables that influence training outcomes before, during, and after training interventions. This process aspect of training effectiveness cannot be viewed in Figure 1 or Table 1. Briefly, the results of this review suggest that self-efficacy, motivation, experience, cognitive ability, and age are characteristics brought to training and are therefore effectiveness variables that influence training outcomes before and during training. Other characteristics that influence training outcomes during training include mastery orientation manipulations, learning principles, and high difficulty. After training, both pretraining and post-training self-efficacy as well as posttraining interventions, learning principles, high difficulty, and a positive transfer environment affect training results.

In sum, although there was not enough research with organizational characteristics and other effectiveness variables to clearly determine a set of key characteristics, this review identified 10 effectiveness variables to consider (see Table 1) when evaluating training programs as they were found to affect training outcomes for a variety of training interventions. Indeed, other effectiveness variables may be equally or more important, depending on the environment and training program. It is acknowledged that the IMTEE does not completely unfold the complexity of training effectiveness. There are underlying interactions and relationships between the effectiveness variables that also affect training outcomes (see Baldwin & Ford, 1988; Cannon-Bowers et al., 1995; Frayne & Geringer, 2000; Holton, 1996; Martocchio, 1994; Quiñones, 1995; Tannenbaum et al., 1993). The following section will discuss the IMTEE as it compares to prior evaluation and effectiveness models.

IMTEE Versus Prior Training Evaluation and Effectiveness Models

Comparing the IMTEE to the four evaluation and five effectiveness models presented earlier in this article, in general, the IMTEE supports all of the models. It was described in the previous sections how the second and third levels of the model efficiently integrate all four evaluation models and how this construction is supported by current research. In addition, the relationships between the evaluation measures were based on current research and theory. However, there is one conflict between the evaluation models regarding the relationship between posttraining attitudes and other training

outcomes. Tannenbaum et al.'s (1993) model suggests that posttraining attitudes are not directly related to any other training outcome. Kraiger's (2002) model proposes that changes in learners, which include posttraining attitudes, are related to organizational payoffs. Kirkpatrick's (1976) model suggests that learning, which also includes posttraining attitudes, is related to transfer performance. This review found that posttraining attitudes are related to cognitive learning, training performance, and transfer performance.

Similar to the IMTEE, the authors of all five training effectiveness models suggested that individual characteristics are related to changes in learners and transfer performance and that training and organizational characteristics are related to transfer performance. However, three models did not present a relationship between training characteristics and transfer performance (Baldwin & Ford, 1988; Holton & Baldwin, 2000; Tannenbaum et al., 1993) and two models did (Broad & Newstrom, 1992; Holton, 1996;). The IMTEE suggests that training characteristics are indeed related to transfer performance. In addition, four effectiveness models (i.e., Baldwin & Ford, 1988; Broad & Newstrom, 1992; Holton, 1996; Holton & Baldwin, 2000) suggest that organizational characteristics are related to changes in learners. The results of the IMTEE do not suggest such a relationship. However, the fifth model (i.e., Tannenbaum et al., 1993) suggests that organizational characteristics are indirectly related to changes in learners through their relationship with particular individual characteristics such as motivation and attitudes. Whether organizational characteristics are directly or indirectly related to changes in learners will need to be further researched before definite conclusions can be made.

Summary and Conclusions

Of the research recommendations made throughout this article, there are three most salient areas in need of further development: reaction measures, motivation, and organizational characteristics. As previously mentioned, new methods for constructing reaction measures are needed so that they provide evaluators with information regarding the appropriateness of training content and design. Toward this end, two current dimensions of reactions to training (i.e., relevance and utility) are feasible starting points. As for motivation, the results of this review suggest that pretraining and posttraining measures of motivation need to be part of training effectiveness research if we are to know how training characteristics increase and/or decrease motivation as well as how changes in motivation affect training outcomes. In addition, recall that a variety of motivational aspects were measured in the literature, making it difficult to accurately determine the effects of motivation in training. Consequently, there is a need for a new, more unified

method of measuring training motivation. As previously mentioned, a viable option is *motivation to improve work through learning*, which is a construct that combines motivation to learn and motivation to transfer (see Naquin & Holton, 2002).

The final of three areas most in need of research attention is organizational characteristics. Consistent with other theoretical works of training effectiveness (e.g., Mathieu & Martineau, 1997; Noe & Colquitt, 2002; Tannenbaum et al., 1993), the results of this review found that more research is needed to determine how specific aspects of organizational contexts influence training outcomes. Therefore, it is difficult to prescribe remedies for creating a learning culture. However, a number of training experts have published promising strategies for creating such a culture that help to identify organizational traits that may be inhibiting positive outcomes (cf. Broad & Newstrom, 1992; Tannenbaum, 1997; Torres & Preskill, 2001).

The IMTEE makes three unique contributions to the training evaluation and effectiveness literatures. First, the IMTEE is the first to be constructed from results of research using a strict set of criteria. As such, it contains only those effectiveness variables found to consistently affect training outcomes. Furthermore, we were able to confirm and advance current training evaluation and effectiveness theories. The approach taken here, however, is not without its limitations. For example, because of the underlying interacting relationships between characteristics, the direct relationships presented in the IMTEE may need to be verified with more sophisticated research techniques (e.g., structural equation modeling). However, because of the small number of studies measuring certain variables, this is not yet feasible.

The IMTEE is also the first to examine relationships between posttraining attitudes and effectiveness variables as well as the remaining evaluation measures. Directing research efforts toward investigations of a variety of posttraining attitudes would further advance our knowledge of the methods that can positively enhance such attitudes as well as their role in influencing training outcomes. The final contribution of the IMTEE is that training effectiveness variables from the past 10 years of research were integrated with all six training evaluation measures, thus providing a current update on training effectiveness research.

In conclusion, the ultimate goal of training is to gain organizational pay-offs. The IMTEE reiterates that changes in learners are necessary for this to occur; however, effectiveness variables are also important and so, ideally, should be part of every training evaluation effort. Thus, the use of a combined model of training evaluation and effectiveness such as the IMTEE may offer a more comprehensive and effective method for evaluating training.

Appendix

Studies Included in the Review

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